

1. Initializing Trend App Development EC2

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name: trend-development-server [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents [Quick Start](#)

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances: 1 [Info](#)

Software Image (AMI)
Canonical, Ubuntu, 26.04, amd64...[read more](#)
ami-0b6d9d3d33ba97d99

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Amazon Machine Image (AMI)

Ubuntu Server 26.04 LTS (HVM), SSD Volume Type
ami-0b6d9d3d33ba97d99 (64-bit (x86)) / ami-0bc7f2dbdccb5303 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Description
Ubuntu Server 26.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).
Canonical, Ubuntu, 26.04, amd64 resolute image

Architecture 64-bit ... **AMI ID** ami-0b6d9d3d33ba97d99 **Creation Date** 2026-06-04 **Username** ubuntu [Verified provider](#)

Instance type [Info](#) [Get advice](#)

Instance type
t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour

☐ All generations [Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

[Cancel](#) [Launch instance](#) [Preview code](#)

aws

Search

[Alt+S]

Ask Amazon Q

1

United States (N. Virginia)

Rathi (620455726130)

root

EC2

Instances

Launch an instance

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

aws-us-east-1

Create new key pair

▼ Network settings Info

Network Info

vpc-0cdfa7c433f29854a

Subnet Info

No preference (Default subnet in any availability zone)

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

We'll create a new security group called 'launch-wizard-5' with the following rules:

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

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aws

Search

[Alt+S]

Ask Amazon Q

1

United States (N. Virginia)

Rathi (620455726130)

root

EC2

Instances

Launch an instance

Create security group

Select existing security group

We'll create a new security group called 'launch-wizard-5' with the following rules:

Allow SSH traffic from

Helps you connect to your instance

Anywhere

0.0.0.0/0

Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

▼ Configure storage Info

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Add new volume

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

▼ Summary

Number of instances Info

1

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd6...read more

ami-0b6d9d3d33ba97d99

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

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2. Installing necessary software packages to run the react application

The image displays two screenshots of a MobaXterm terminal window. The top screenshot shows the initial setup of nvm and Node.js. The terminal text is as follows:

```
ubuntu@ip-172-31-8-34:~$ # Download and install nvm:
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.6/install.sh | bash

# in lieu of restarting the shell
\ . "$HOME/.nvm/nvm.sh"

# Download and install Node.js:
nvm install 24

# Verify the Node.js version:
node -v # Should print "v24.19.0".

# Verify npm version:
npm -v # Should print "11.17.0".
```

The bottom screenshot shows the installation of the 'serve' package using npm. The terminal text is as follows:

```
ubuntu@ip-172-31-8-34:~$ npm install -g serve

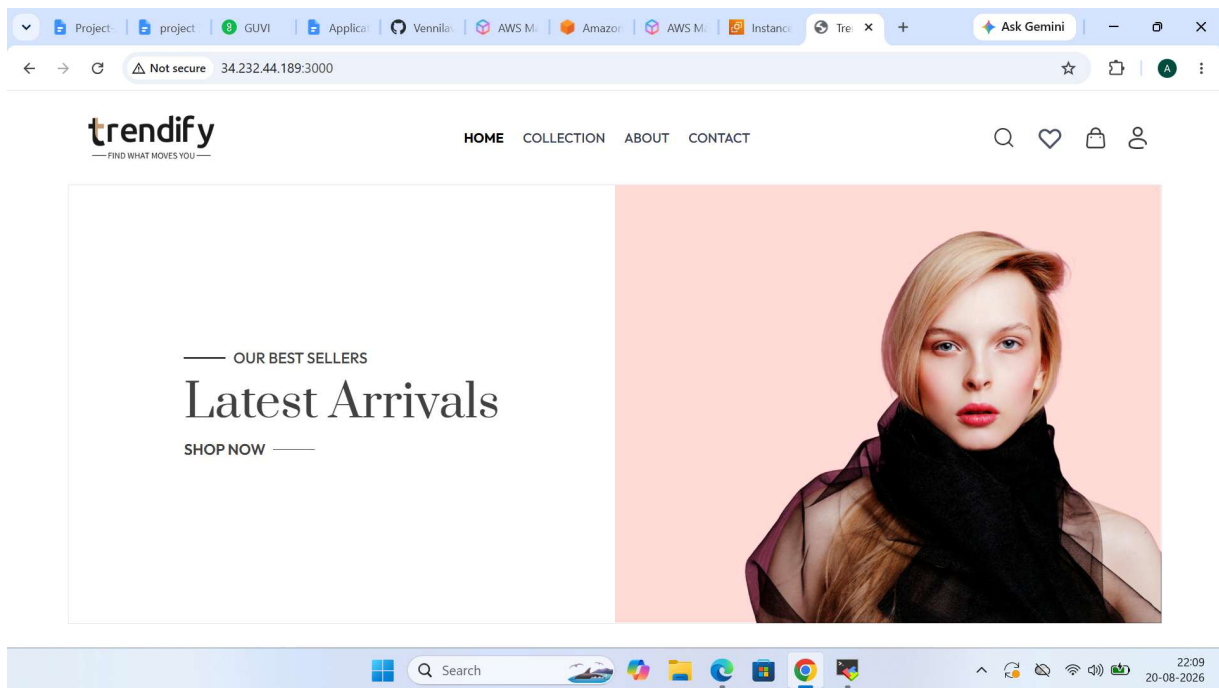
added 85 packages in 4s

26 packages are looking for funding
  run `npm fund` for details

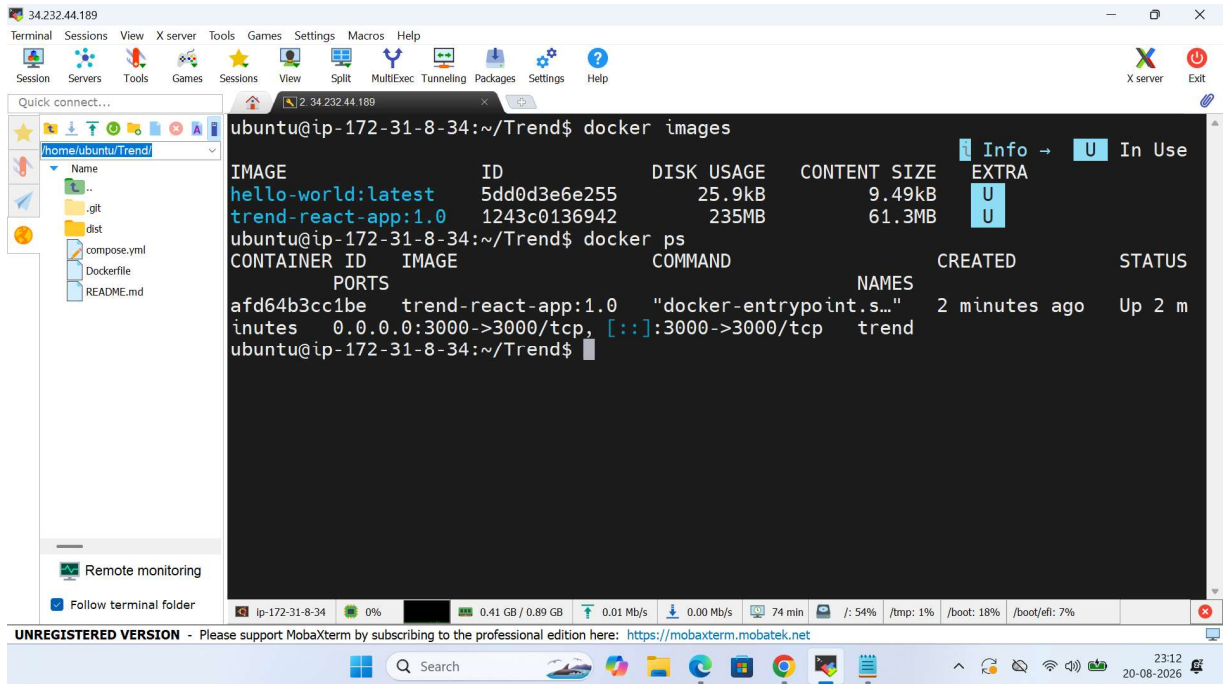
npm notice
npm notice New major version of npm available! 11.17.0 -> 12.0.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.0.2
npm notice To update run: npm install -g npm@12.0.2
npm notice
ubuntu@ip-172-31-8-34:~$
```

Both screenshots show the MobaXterm interface with a file explorer on the left and a status bar at the bottom. The status bar indicates the terminal is connected to ip-172-31-8-34 and shows various system metrics.

3. Run the website



4. Dockerize the website



The screenshot shows a MobaXterm terminal window with a file explorer on the left and a terminal window on the right. The terminal window is titled '34.232.44.189' and shows the following commands and output:

```
ubuntu@ip-172-31-8-34:~/Trend$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	U
trend-react-app:1.0	1243c0136942	235MB	61.3MB	U

```
ubuntu@ip-172-31-8-34:~/Trend$ docker ps
```

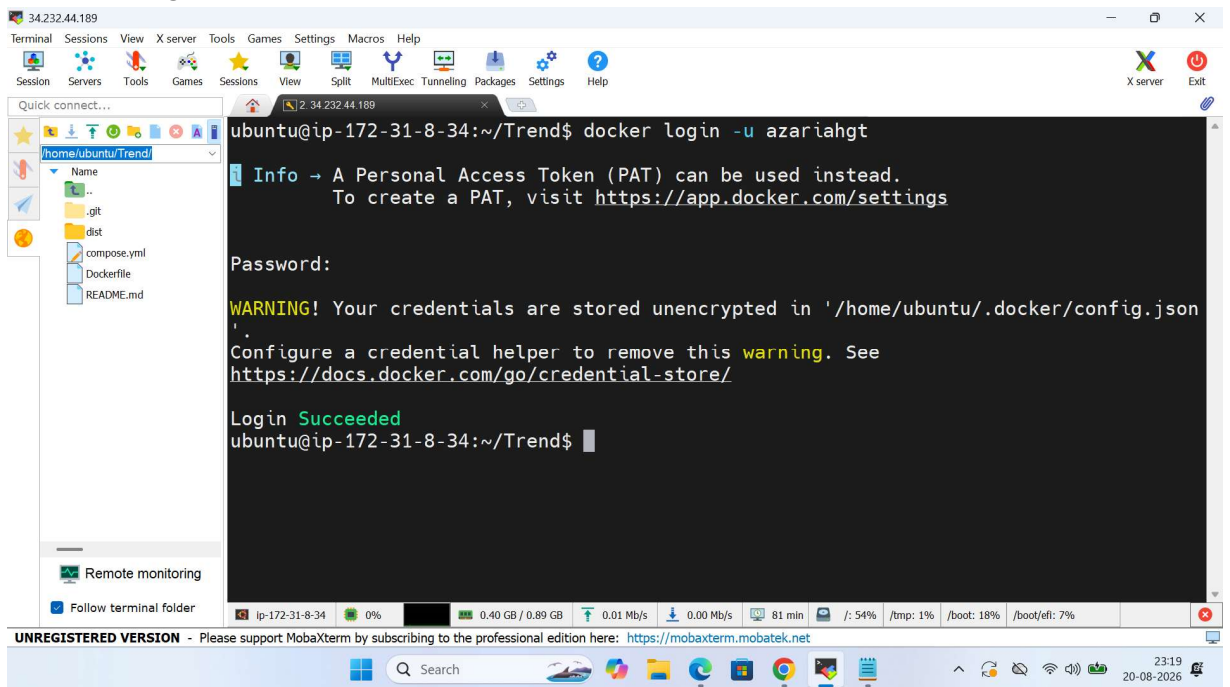
CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
afdf64b3cc1be	trend-react-app:1.0	"docker-entrypoint.s..."	2 minutes ago	Up 2 m

```
inutes 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp trend
ubuntu@ip-172-31-8-34:~/Trend$
```

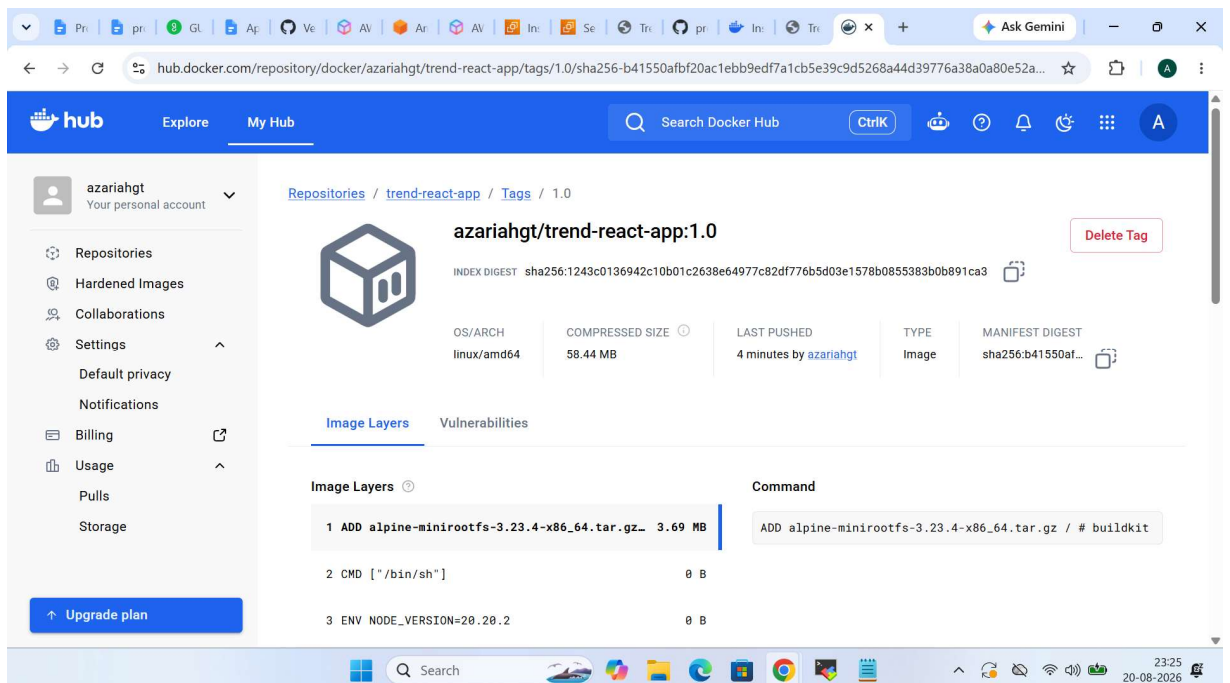
The terminal window also shows a file explorer on the left with the following files: Name, .git, dist, compose.yml, Dockerfile, and README.md. The terminal window is titled '34.232.44.189' and shows the following commands and output:

UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>

5. Dockerhub Login



6. Dockerhub Repo

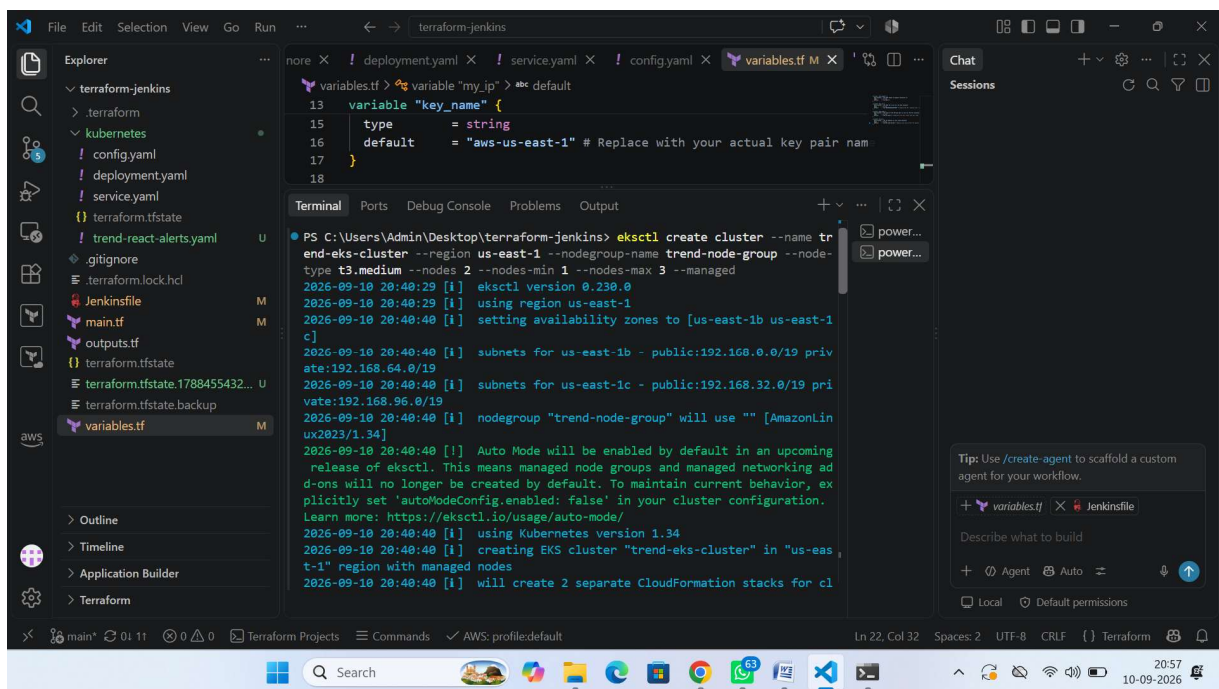


7.EKS Cluster Creation Commands [Files are in the Github repo]

Run from the windows powershell / visual code pc

Step1:

`eksctl create cluster --name trend-eks-cluster --region us-east-1 --nodegroup-name trend-node-group --node-type t3.medium --nodes 2 --nodes-min 1 --nodes-max 3 --managed`



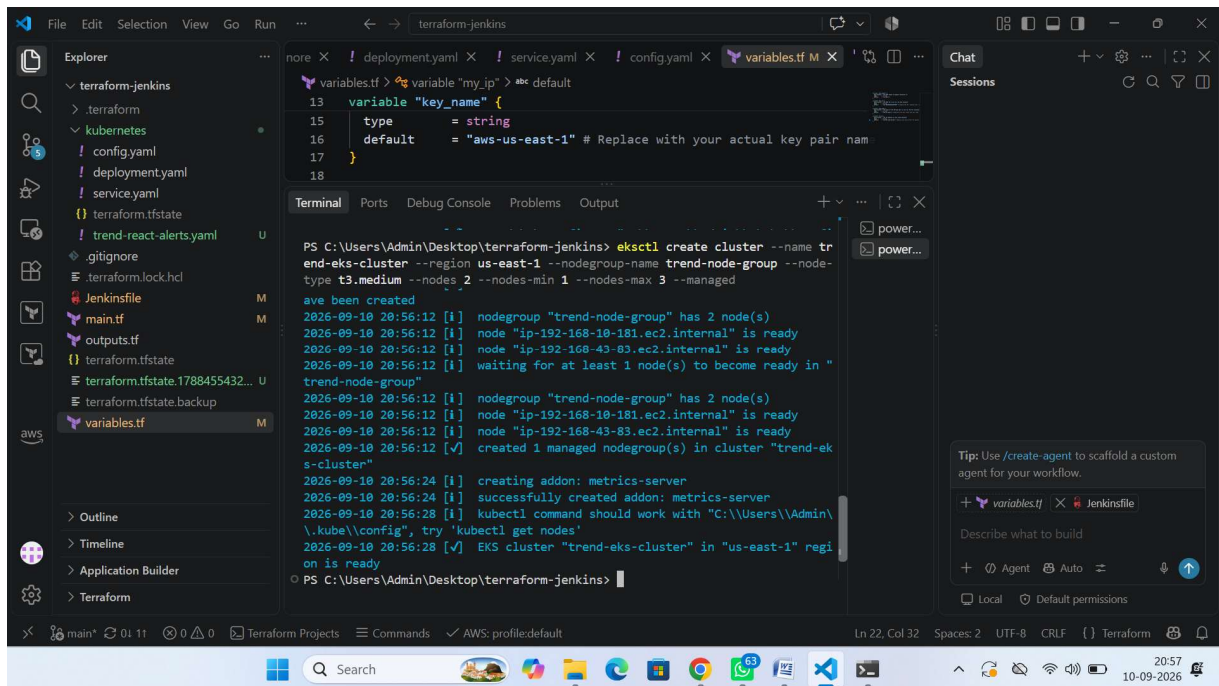
The screenshot shows the Visual Studio Code interface with a file explorer on the left, a code editor in the center, and a terminal window at the bottom. The file explorer shows a project named 'terraform-jenkins' with subfolders for 'terraform', 'kubernetes', and 'aws'. The code editor displays a file named 'variables.tf' with the following content:

```
13 variable "my_ip" {
14     type = string
15     default = "aws-us-east-1" # Replace with your actual key pair name
16 }
17
18
```

The terminal window shows the execution of the command `eksctl create cluster --name trend-eks-cluster --region us-east-1 --nodegroup-name trend-node-group --node-type t3.medium --nodes 2 --nodes-min 1 --nodes-max 3 --managed`. The output shows the following steps:

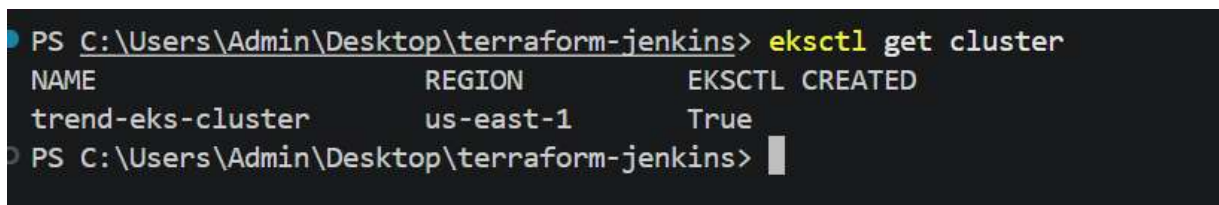
- eksctl version 0.230.0
- using region us-east-1
- setting availability zones to [us-east-1b us-east-1c]
- subnets for us-east-1b - public:192.168.0.0/19 private:192.168.64.0/19
- subnets for us-east-1c - public:192.168.32.0/19 private:192.168.96.0/19
- nodegroup "trend-node-group" will use "" [AmazonLinux2023/1.34]
- Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no longer be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: <https://eksctl.io/usage/auto-mode/>
- using Kubernetes version 1.34
- creating EKS cluster "trend-eks-cluster" in "us-east-1" region with managed nodes
- will create 2 separate CloudFormation stacks for cl

The terminal window also shows a chat window on the right with a tip: "Tip: Use /create-agent to scaffold a custom agent for your workflow." and a search bar at the bottom.



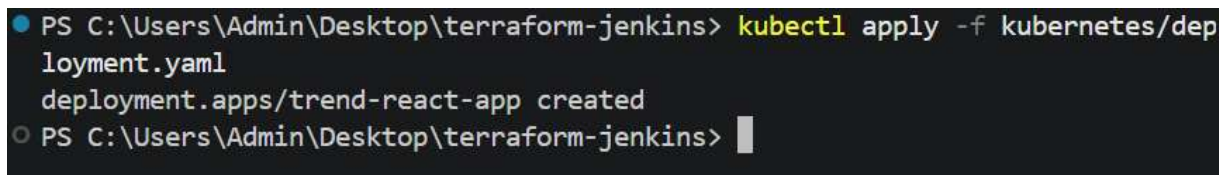
Step2:

eksctl get cluster



Step 3:

kubectl apply -f kubernetes/deployment.yaml



Step 4:

kubectl apply -f kubernetes/service.yaml

```
PS C:\Users\Admin\Desktop\terraform-jenkins> kubectl apply -f kubernetes/service.yaml
service/trend-react-service created
```

Step5:

kubectl get deployments

```
PS C:\Users\Admin\Desktop\terraform-jenkins> kubectl get deployments
NAME                                READY    UP-TO-DATE    AVAILABLE
trend-react-app                    2/2      2              2
```

kubectl get pods

```
PS C:\Users\Admin\Desktop\terraform-jenkins> kubectl get pods
NAME                                READY    STATUS
trend-react-app-7f98dd4b7c-rj45g    1/1      Running
trend-react-app-7f98dd4b7c-vnpq4    1/1      Running
```

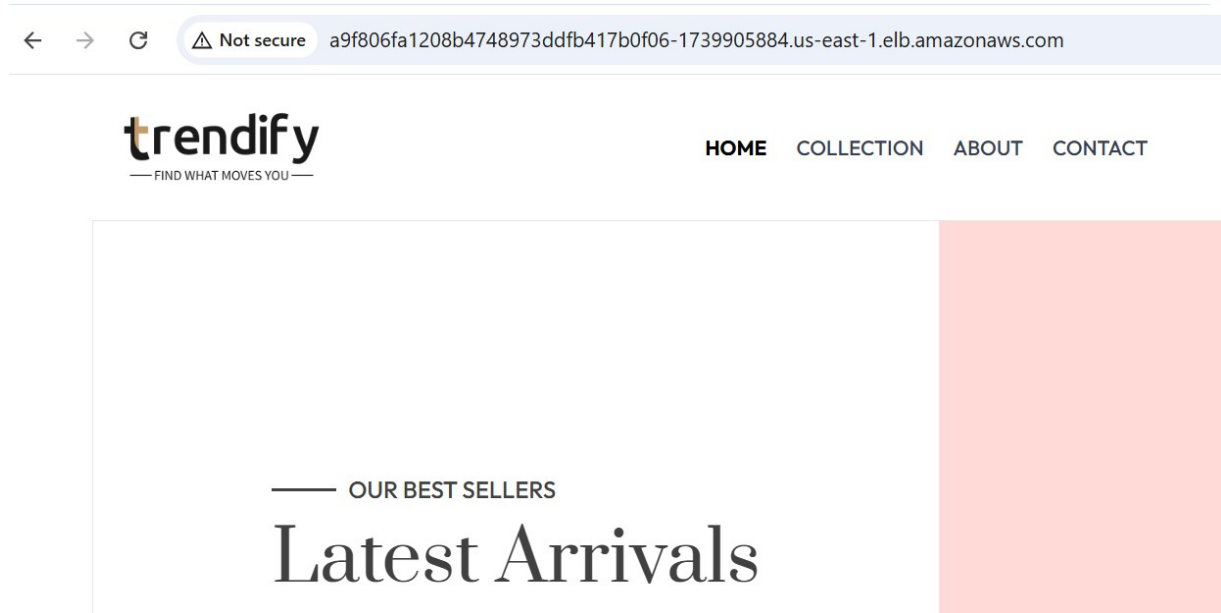
kubectl get svc

```
PS C:\Users\Admin\Desktop\terraform-jenkins> kubectl get svc
NAME                                TYPE                CLUSTER-IP
trend-react-service                LoadBalancer        10.100.253.38
kubernetes                         ClusterIP            10.100.0.1
```

Step 8:

Verify load balancer url: Open the `EXTERNAL-IP/address` address in your browser.

<http://a9f806fa1208b4748973ddfb417b0f06-1739905884.us-east-1.elb.amazonaws.com/>



Provisioning EC2 preinstalled with Jenkins and other necessary packages through Terraform [files are in github repo]

```
● PS C:\Users\Admin\Desktop\terraform-jenkins> terraform apply

Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

# aws_eks_access_entry.jenkins_access will be created
+ resource "aws_eks_access_entry" "jenkins_access" {
  + access_entry_arn = (known after apply)
  + cluster_name     = "trend-eks-cluster"
  + created_at       = (known after apply)
  + id               = (known after apply)
  + kubernetes_groups = (known after apply)
  + modified_at      = (known after apply)
  + principal_arn     = (known after apply)
  + region           = "us-east-1"
  + tags_all         = (known after apply)
  + type             = "STANDARD"
  + user_name        = (known after apply)
}
```

Apply complete! Resources: 17 added, 0 changed, 0 destroyed.

Outputs:

```
jenkins_public_ip = "54.152.172.187"
jenkins_url = "http://54.152.172.187:8080"
subnet_id = "subnet-086cf789813d40c52"
vpc_id = "vpc-0deb23cc70afb3108"
PS C:\Users\Admin\Desktop\terraform-jenkins> █
```

in the jenkins installed server

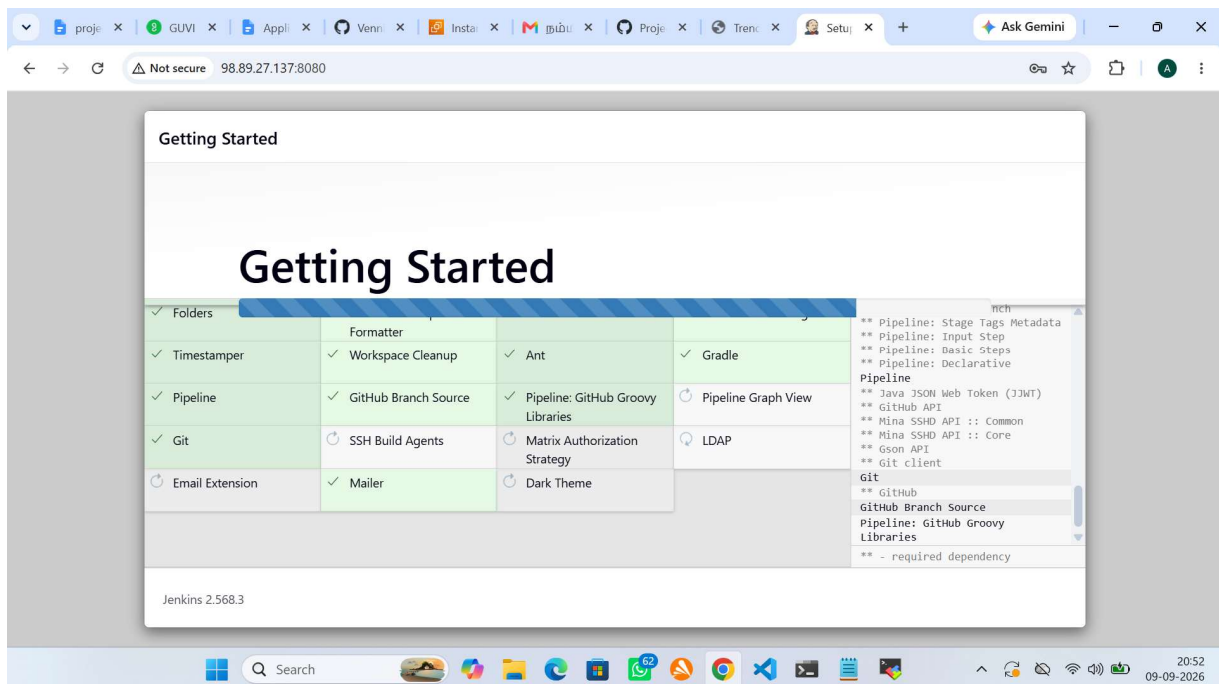
aws eks update-kubeconfig --region us-east-1 --name trend-eks-cluster

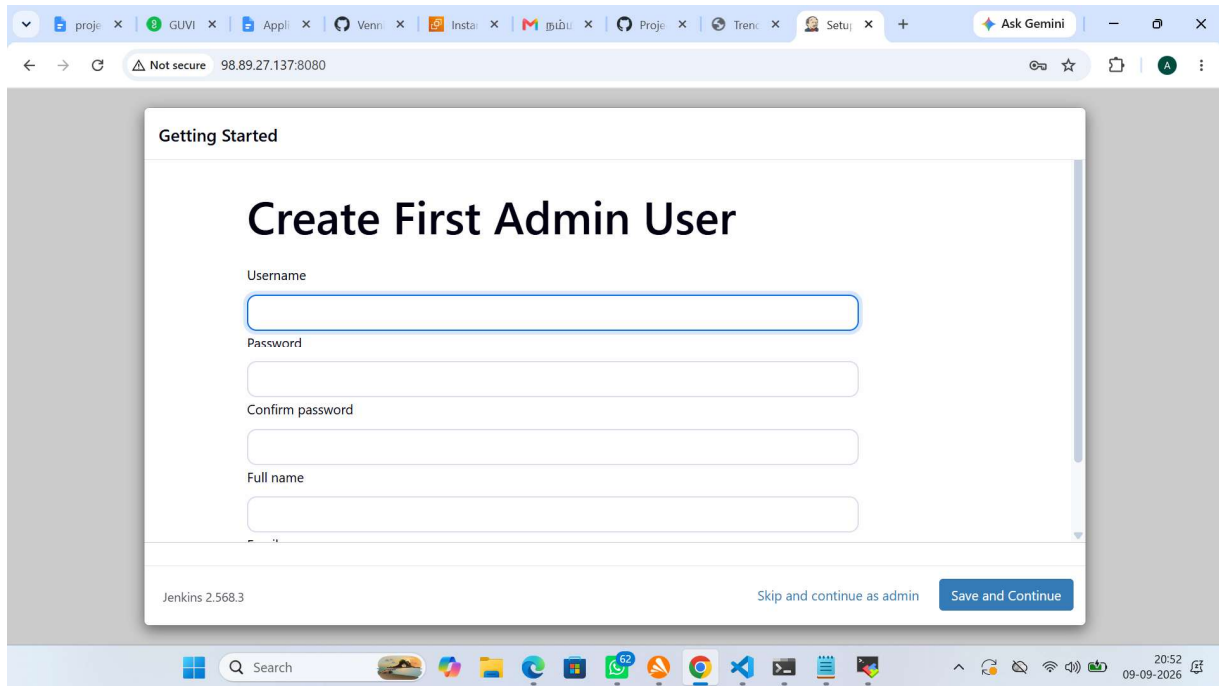
```
ubuntu@ip-10-0-1-11:~$ aws eks update-kubeconfig --region us-east-1 --name trend-eks-cluster
kubectl get nodes
Added new context arn:aws:eks:us-east-1:620455726130:cluster/trend-eks-cluster to /home/ubuntu/.
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-10-181.ec2.internal      Ready    <none>   25m    v1.34.10-eks-cb19647
ip-192-168-43-83.ec2.internal       Ready    <none>   25m    v1.34.10-eks-cb19647
ubuntu@ip-10-0-1-11:~$ █
```

```
kubectl get nodes
```

```
ubuntu@ip-10-0-1-11:~$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
trend-react-app-7f98dd4b7c-rj45g   1/1     Running   0           19m
trend-react-app-7f98dd4b7c-wnpq4   1/1     Running   0           19m
ubuntu@ip-10-0-1-11:~$
```

Jenkins Setup steps





Installing necessary plugins

Pipeline Stage View
GitHub Pipe Line for Blue ocean
Docker Pipeline
Kubernetes
Email Extension

Browser tabs: proje, GUVI, Appli, Venn, Sequ, M, Webli, Trenc, Plugi, Ask Gemini

Address bar: Not secure 98.89.27.137:8080/manage/pluginManager/available

Jenkins / Manage Jenkins / Plugins

Search: Email

Buttons: Install, Refresh

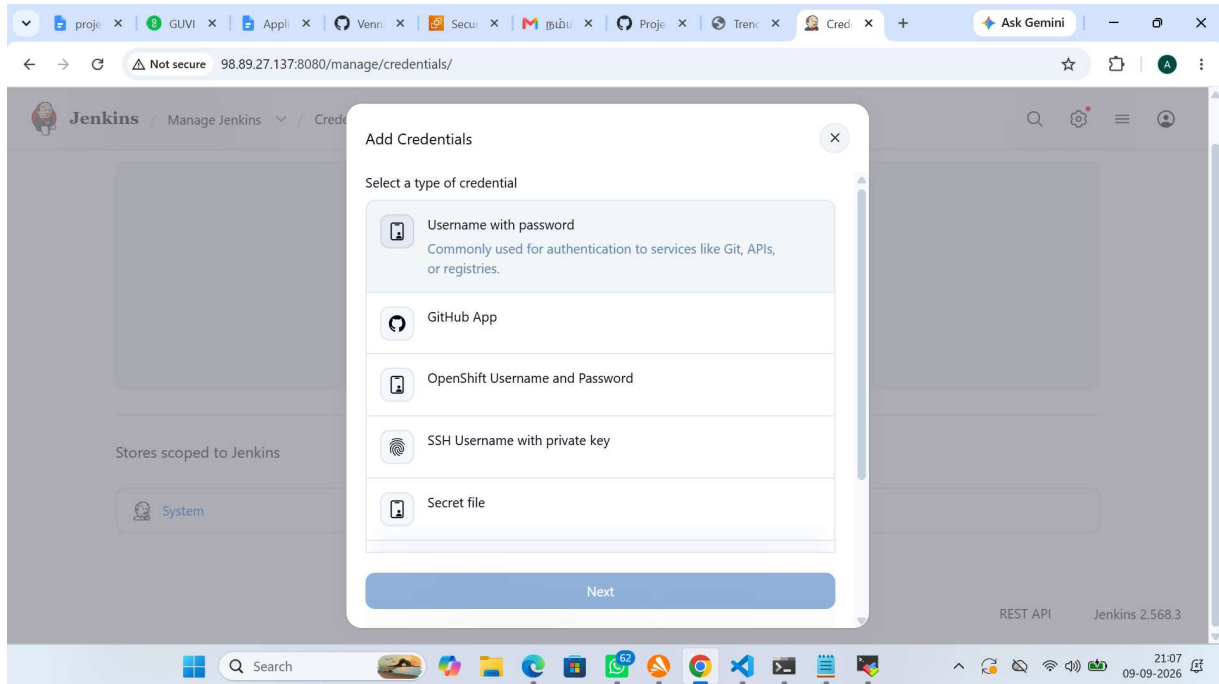
Left sidebar:

- Updates
- Available plugins
- Installed plugins
- Advanced settings
- Download progress

Install	Name	Released	Health
☒	Pipeline: Stage View 2.41 User Interface Pipeline Stage View Plugin.	4 mo 23 days ago	100
☒	GitHub Pipeline for Blue Ocean 1.27.26 External Site/Tool Integrations User Interface BlueOcean GitHub organization pipeline creator This plugin is deprecated. In general, this means that it is either obsolete, no longer being developed, or may no longer work. Learn more.	1 mo 2 days ago	97
☒	Kubernetes 4547.v52f3080db_8cd Cloud Providers Cluster Management kubernetes Agent Management This plugin integrates Jenkins with Kubernetes	27 days ago	97

Windows taskbar: Search, 20:58 09-09-2026

Setting up Global Credentials:



pro | GUV | App | Ven | AW | AW | Inst | Wel | a6b | 3.9 | aza | Tre | Ask Gemini

Not secure 3.95.59.12:8080/manage/credentials/

Jenkins

Manage Jenkins / Credentials

← Add Username with password

Scope ?
Global (Jenkins, nodes, items, all child items, etc)

Username
azariahgt

☐ Treat username as secret ?

Password
.....

ID ?
dockerhub-credentials

Create

Stores scoped to Jenkins

System

21:09 02-09-2026

pro | GUV | App | Ven | AW | AW | Inst | Wel | a6b | 3.9 | aza | Tre | Ask Gemini

Not secure 3.95.59.12:8080/manage/credentials/

Jenkins

Manage Jenkins / Credentials

+ Add Credentials

?

 dockerhub-credentials	azariahgt/*****	 
System - Global - dockerhub-credentials-details		
 github-credentials	azariah.gt@gmail.com/*****	 
System - Global		

Stores scoped to Jenkins

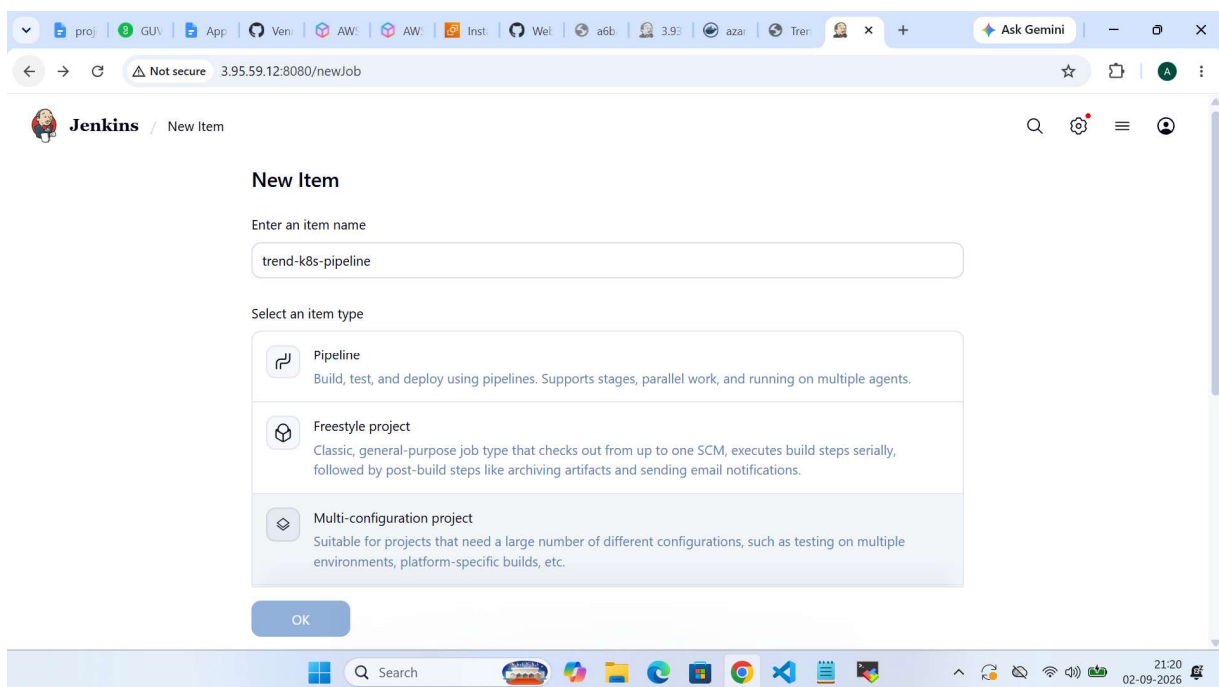
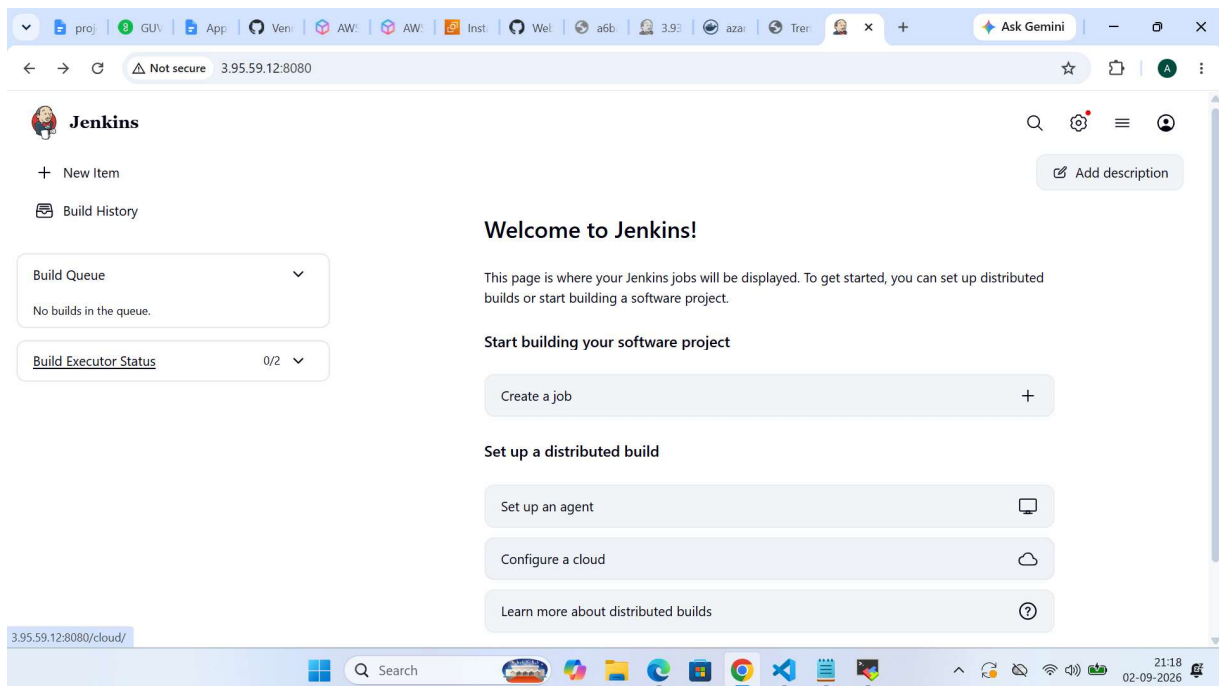
 System

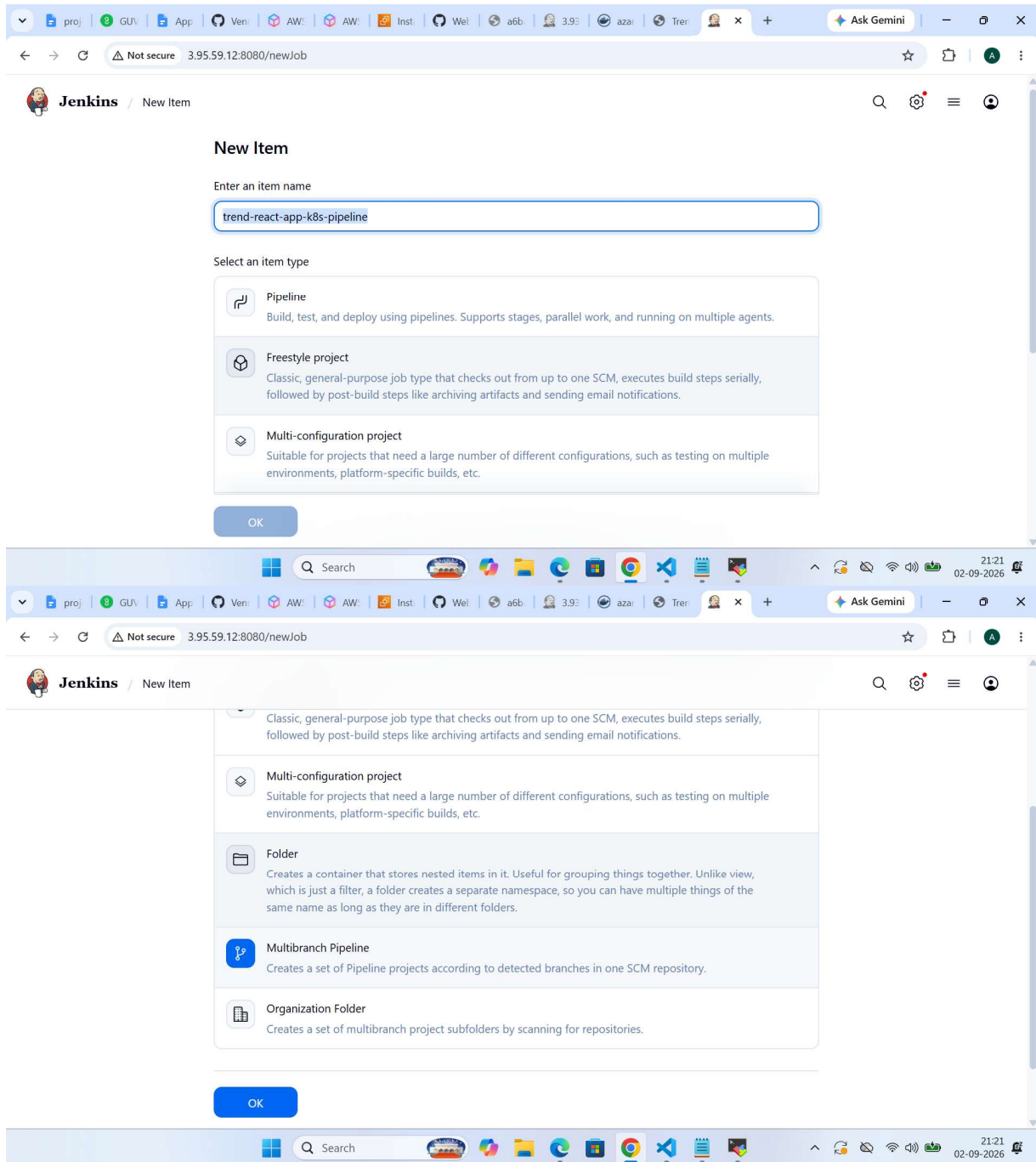
Domains: Global

REST API Jenkins 2.568.3

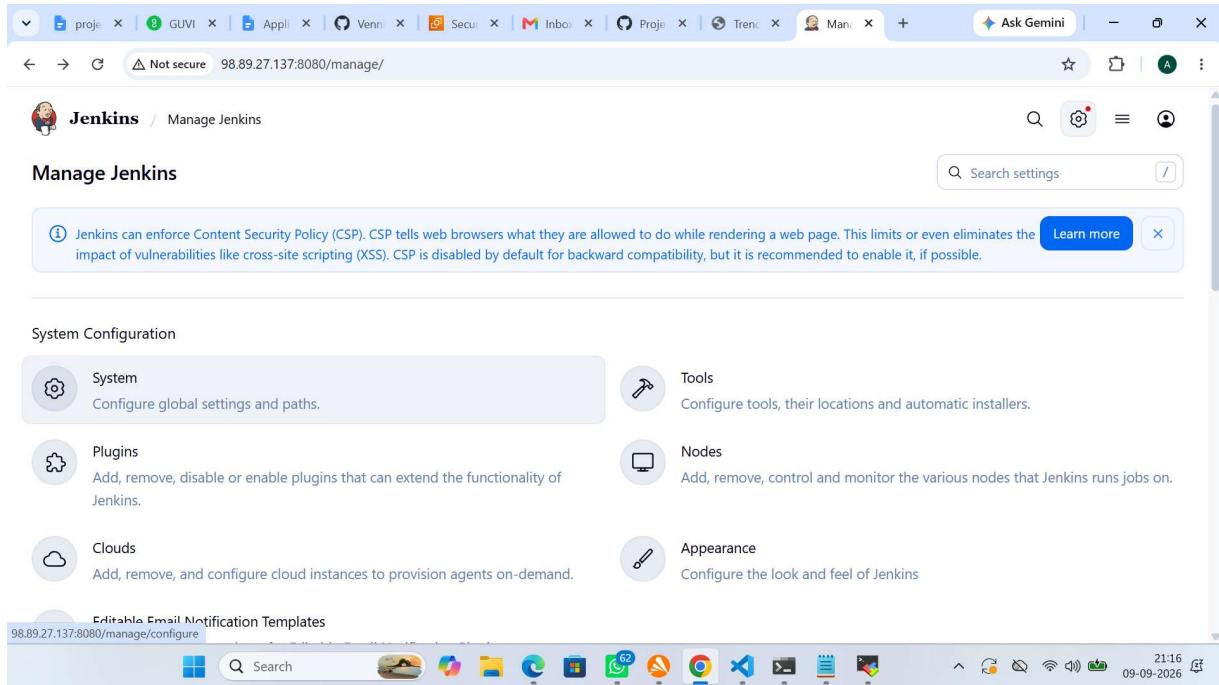
21:11 02-09-2026

Jenkins Job Creation:

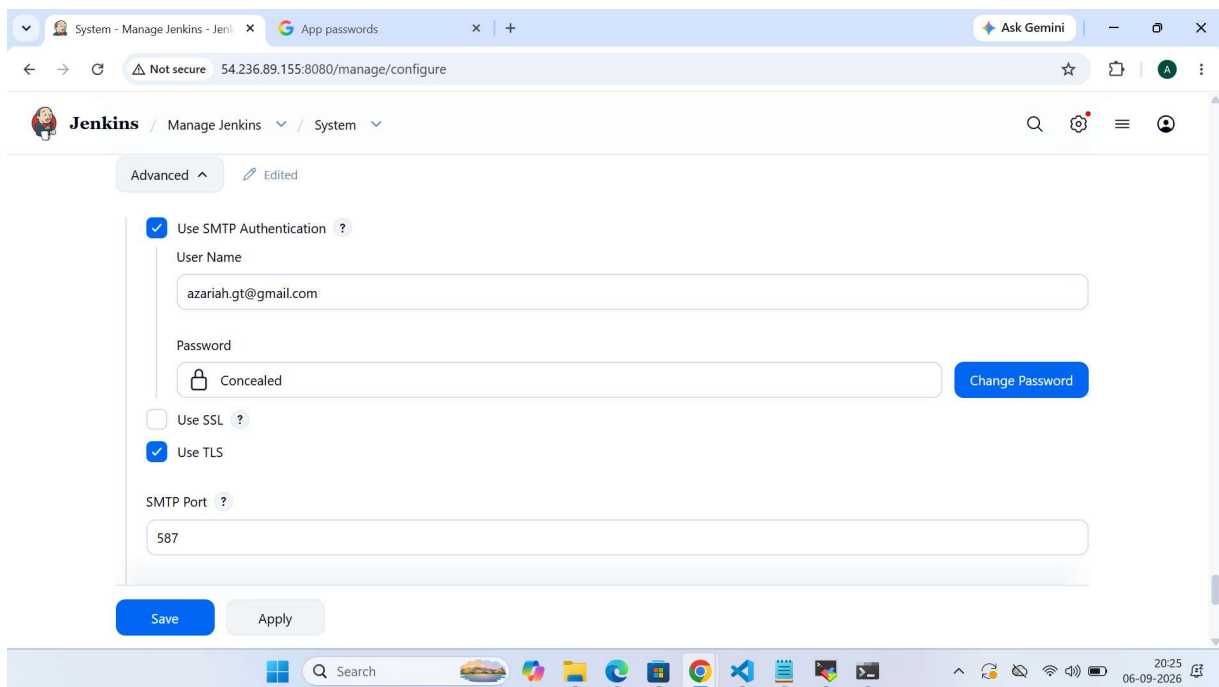




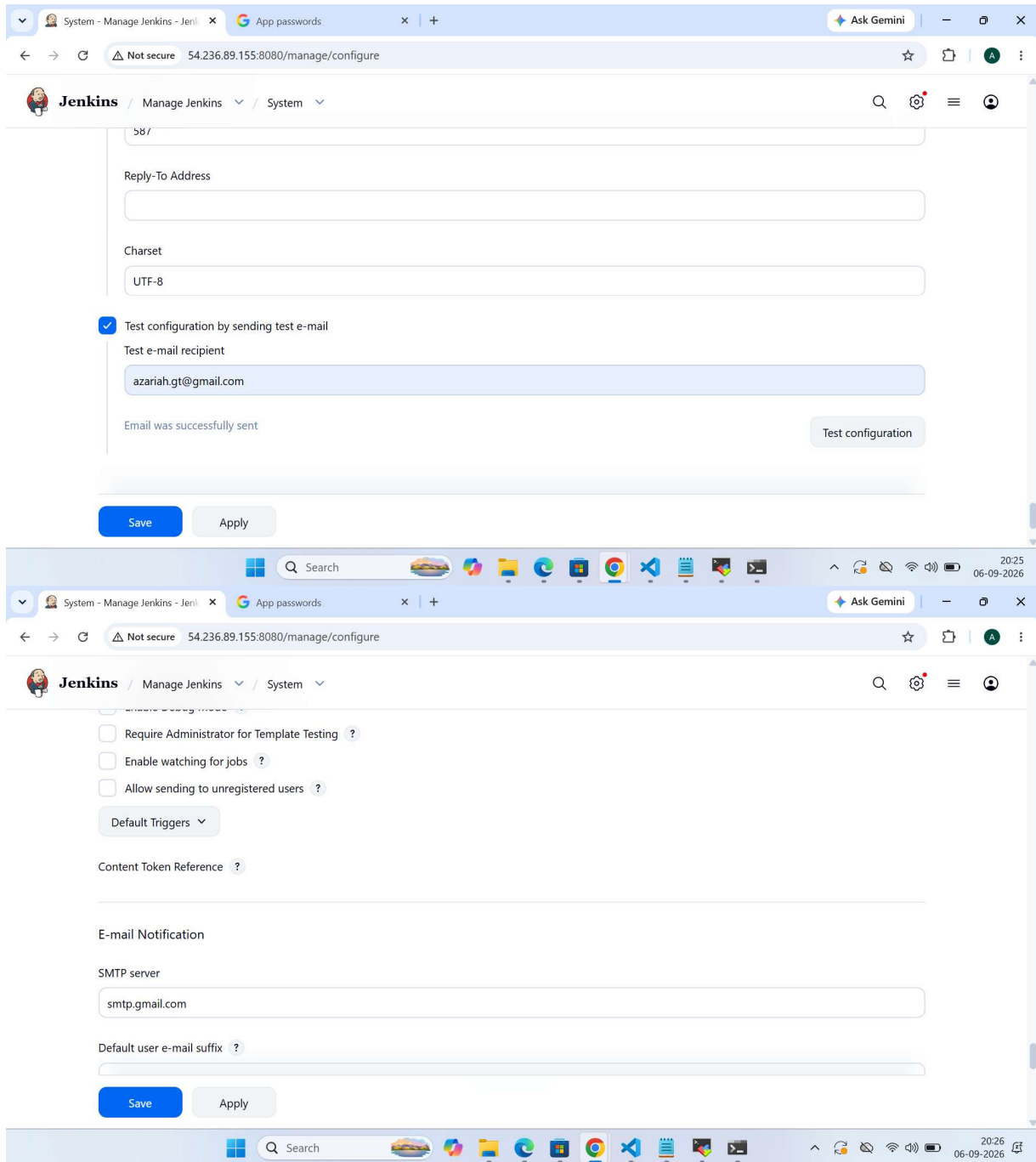
Configuring Email setup

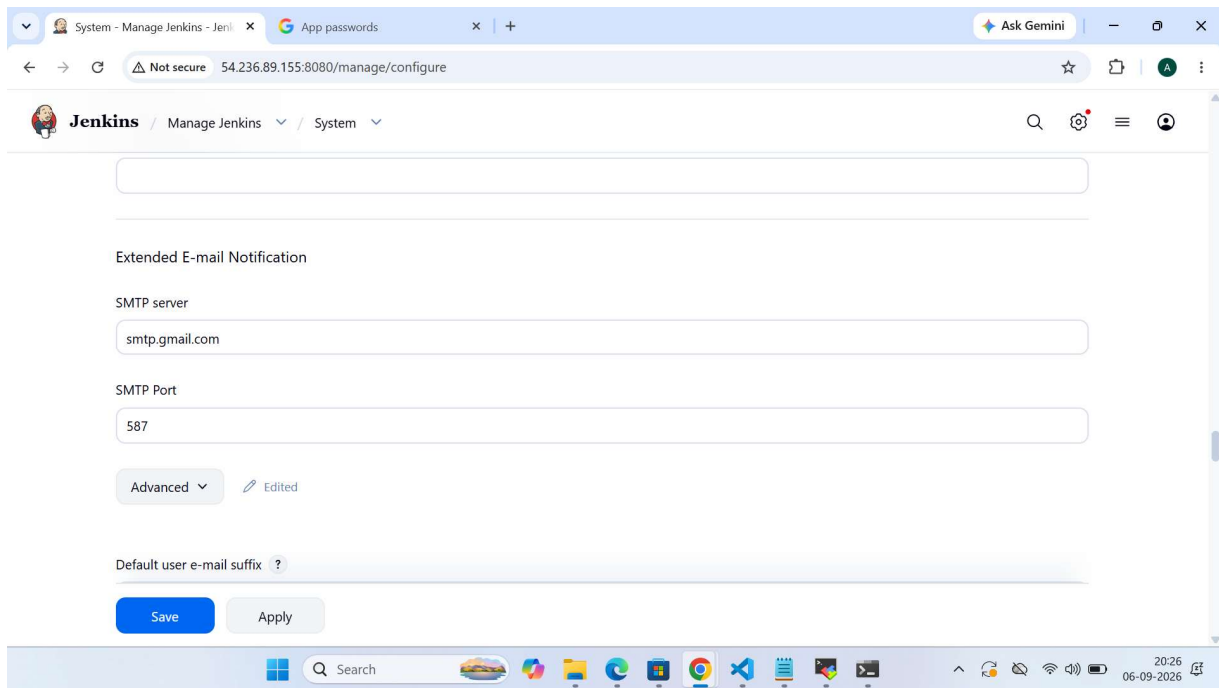


The screenshot shows the Jenkins 'Manage Jenkins' interface in a web browser. The browser's address bar displays '98.89.27.137:8080/manage/'. The page title is 'Jenkins / Manage Jenkins'. A search bar is located in the top right corner. A blue informational banner at the top states: 'Jenkins can enforce Content Security Policy (CSP). CSP tells web browsers what they are allowed to do while rendering a web page. This limits or even eliminates the impact of vulnerabilities like cross-site scripting (XSS). CSP is disabled by default for backward compatibility, but it is recommended to enable it, if possible.' Below this banner, the 'System Configuration' section is visible, featuring several icons and links: 'System' (Configure global settings and paths), 'Tools' (Configure tools, their locations and automatic installers), 'Plugins' (Add, remove, disable or enable plugins that can extend the functionality of Jenkins), 'Nodes' (Add, remove, control and monitor the various nodes that Jenkins runs jobs on), 'Clouds' (Add, remove, and configure cloud instances to provision agents on-demand), and 'Appearance' (Configure the look and feel of Jenkins). At the bottom of the configuration section, there is a link for 'Editable Email Notification Templates'.

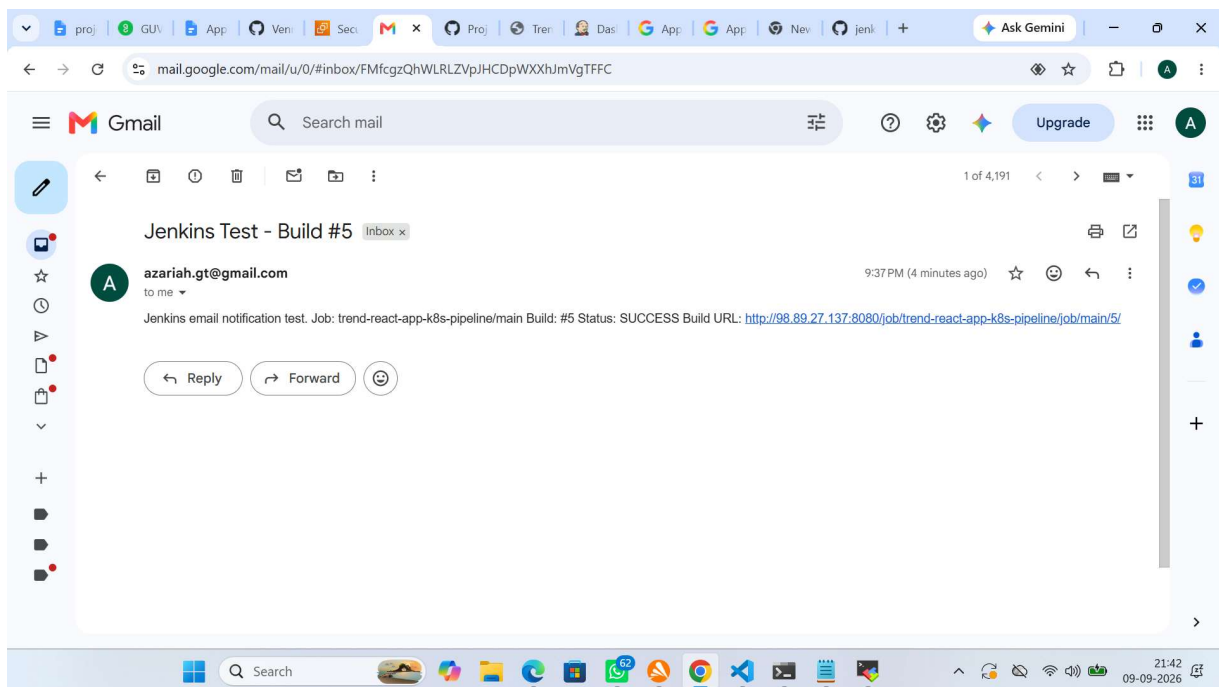


The screenshot shows the 'System' configuration page within the Jenkins 'Manage Jenkins' interface. The browser's address bar displays '54.236.89.155:8080/manage/configure'. The page title is 'Jenkins / Manage Jenkins / System'. The configuration is presented in a form with a 'Save' button and an 'Apply' button. The 'Advanced' section is expanded, showing the following settings: 'Use SMTP Authentication' is checked, with a 'User Name' field containing 'azariah.gt@gmail.com' and a 'Password' field labeled 'Concealed' with a 'Change Password' button; 'Use SSL' is unchecked; 'Use TLS' is checked; and the 'SMTP Port' is set to '587'.





Build Complete Status Email received



Monitoring server installation through helm [kubernetes]

1. Installing helm

```
HELM_BUILDKITE_APT_KEY_ID="DDF78C3E6EBB2D2CC223C95C62BA89D07698DBC6"
```

```
sudo apt-get install curl gpg apt-transport-https --yes
```

```
curl -fsSL https://packages.buildkite.com/helm-linux/helm-debian/gpgkey > "${TMPDIR:-/tmp}/helm.gpg"
```

```
# Ensure that the key ID matches to prevent a repository compromise from establishing an  
attacker controlled key
```

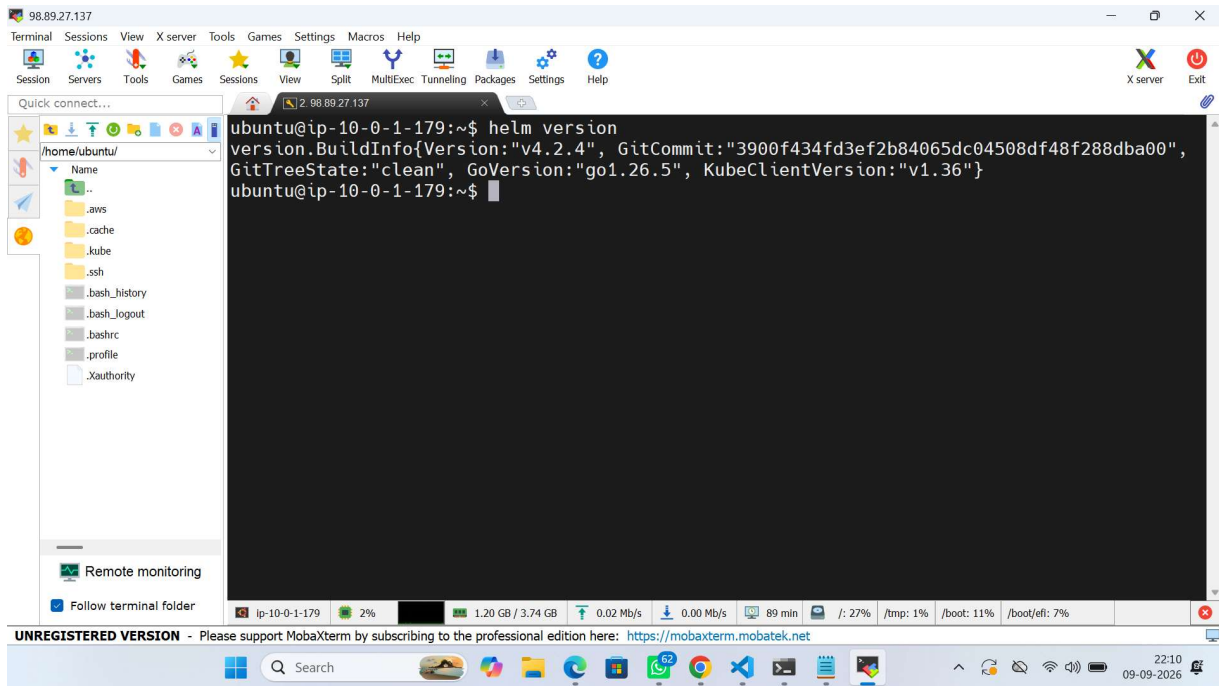
```
if [ "$(gpg --show-keys --with-colons "${TMPDIR:-/tmp}/helm.gpg" | awk -F: '$1 == "fpr" {print  
$10}' | head -n 1)" != "${HELM_BUILDKITE_APT_KEY_ID}" ]; then echo "ERROR: Unexpected  
Helm APT key ID: potential key compromise"; exit 1; fi
```

```
cat "${TMPDIR:-/tmp}/helm.gpg" | gpg --dearmor | sudo tee /usr/share/keyrings/helm.gpg >  
/dev/null
```

```
echo "deb [signed-by=/usr/share/keyrings/helm.gpg] https://packages.buildkite.com/helm-  
linux/helm-debian/any/ any main" | sudo tee /etc/apt/sources.list.d/helm-stable-debian.list
```

```
sudo apt-get update
```

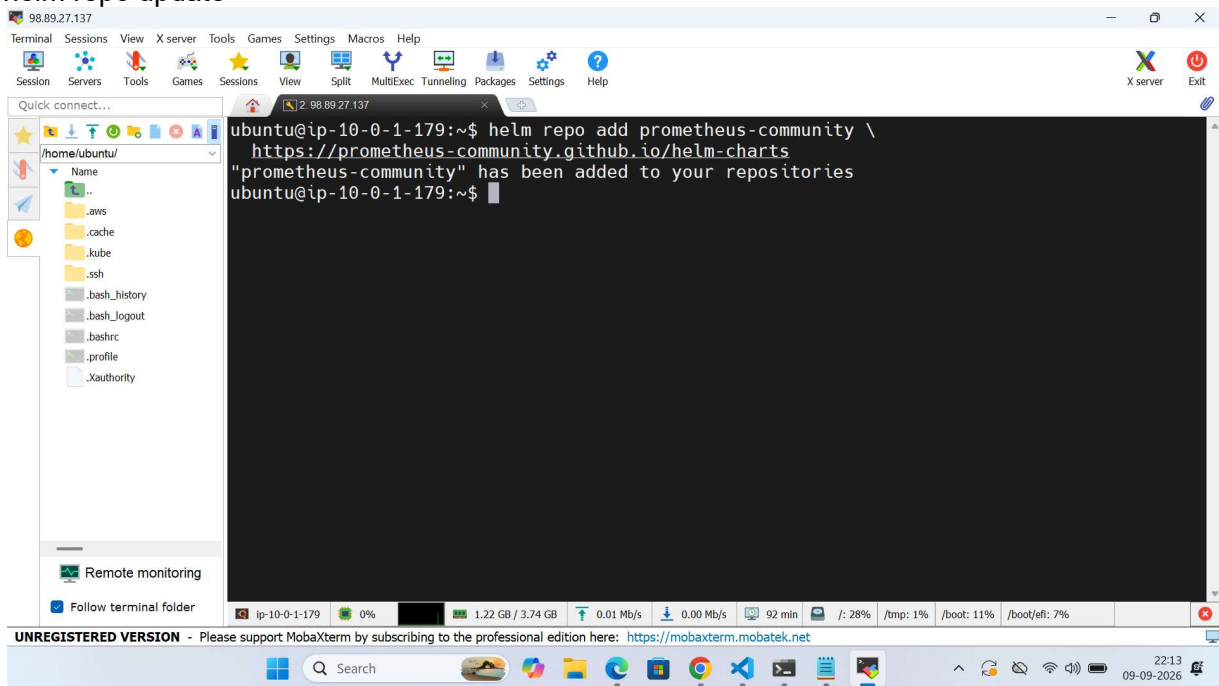
```
sudo apt-get install helm
```

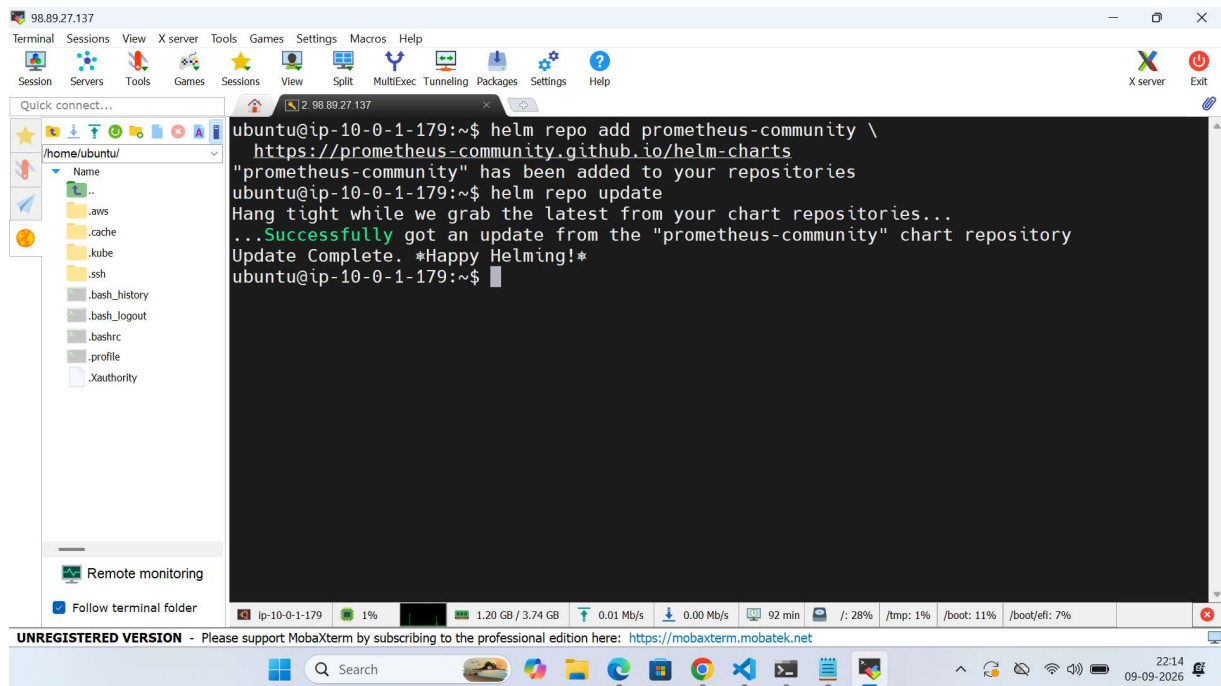


Add the prometheus Helm repository

```
helm repo add prometheus-community \
https://prometheus-community.github.io/helm-charts
```

```
helm repo update
```

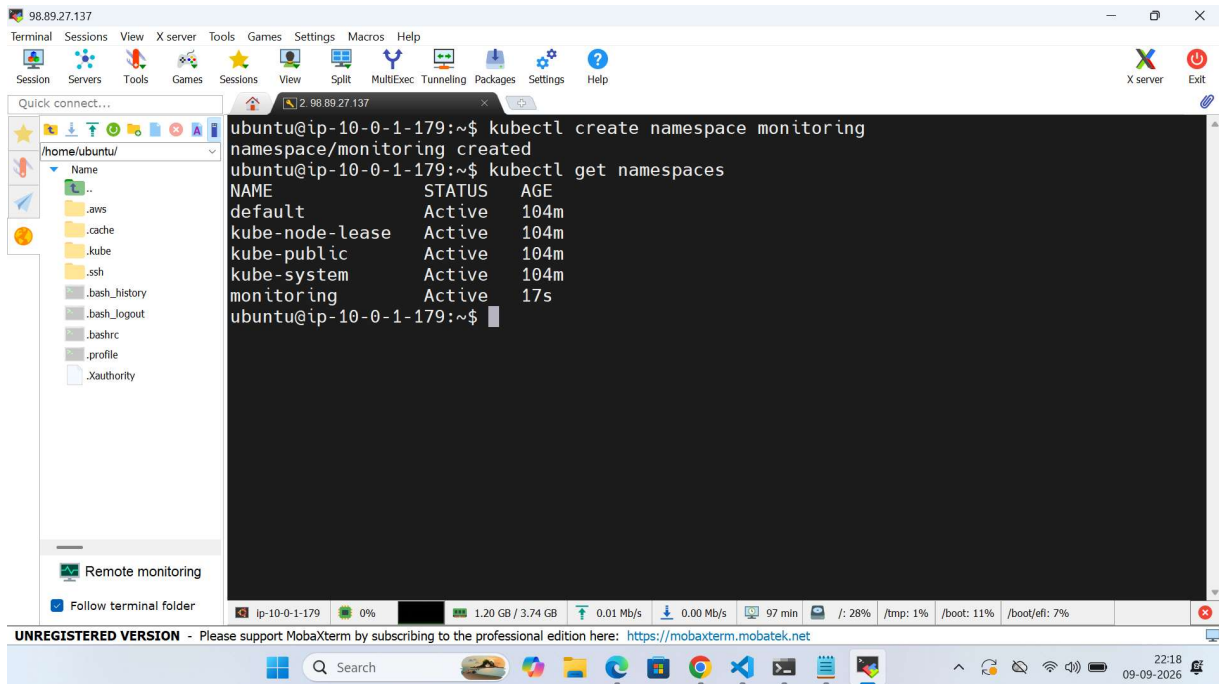




Create monitoring namespace

kubectl create namespace monitoring

kubectl create namespace monitoring

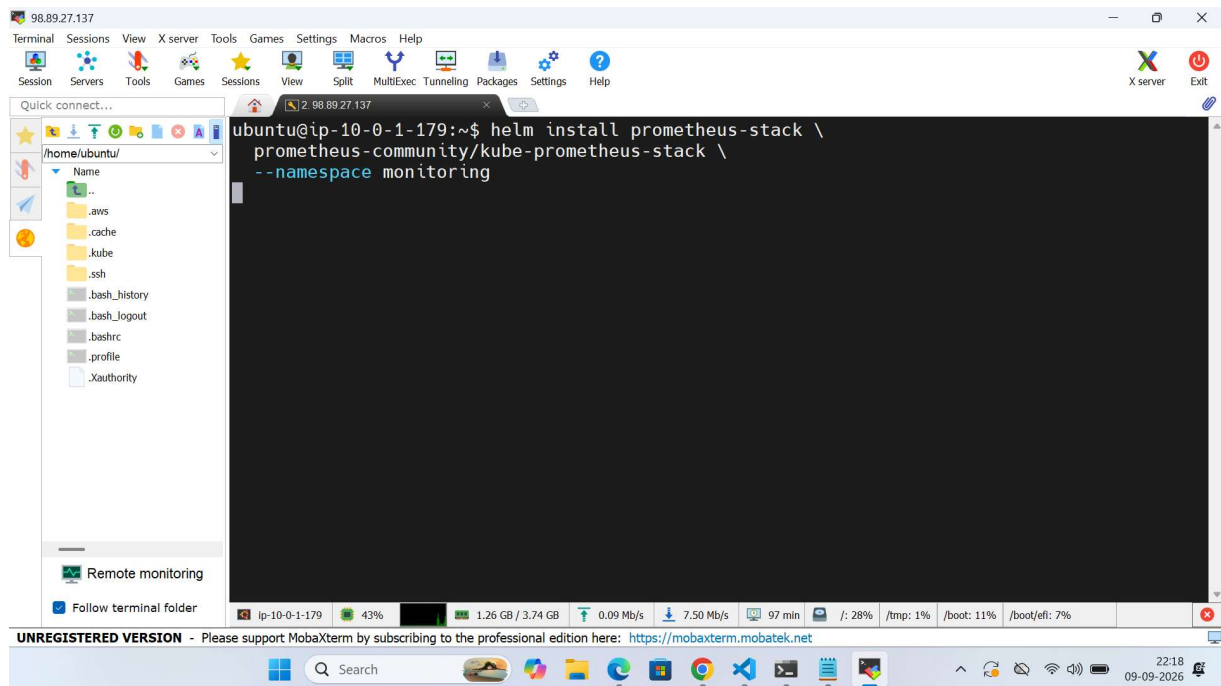


Install kube-prometheus-stack

```
helm install prometheus-stack \
  prometheus-community/kube-prometheus-stack \
  --namespace monitoring
```

Output will be as follows

```
NAME: prometheus-stack
LAST DEPLOYED: Wed Sep 9 16:45:00 2026
NAMESPACE: monitoring
STATUS: deployed
REVISION: 1
DESCRIPTION: Install complete
TEST SUITE: None
NOTES:
kube-prometheus-stack has been installed. Check its status by running:
  kubectl --namespace monitoring get pods -l "release=prometheus-stack"
```



Get Grafana 'admin' user password by running:

```
kubectl --namespace monitoring get secrets prometheus-stack-grafana -o
jsonpath="{.data.admin-password}" | base64 -d ; echo
```

Access Grafana local instance:

```
export POD_NAME=$(kubectl --namespace monitoring get pod -l
"app.kubernetes.io/name=grafana,app.kubernetes.io/instance=prometheus-stack" -oname)
kubectl --namespace monitoring port-forward $POD_NAME 3000
```

Get your grafana admin user password by running:

```
kubectl get secret --namespace monitoring -l app.kubernetes.io/component=admin-secret -o
jsonpath="{.items[0].data.admin-password}" | base64 --decode ; echo
```

Access grafana

```
kubectl get svc -n monitoring | grep grafana
```

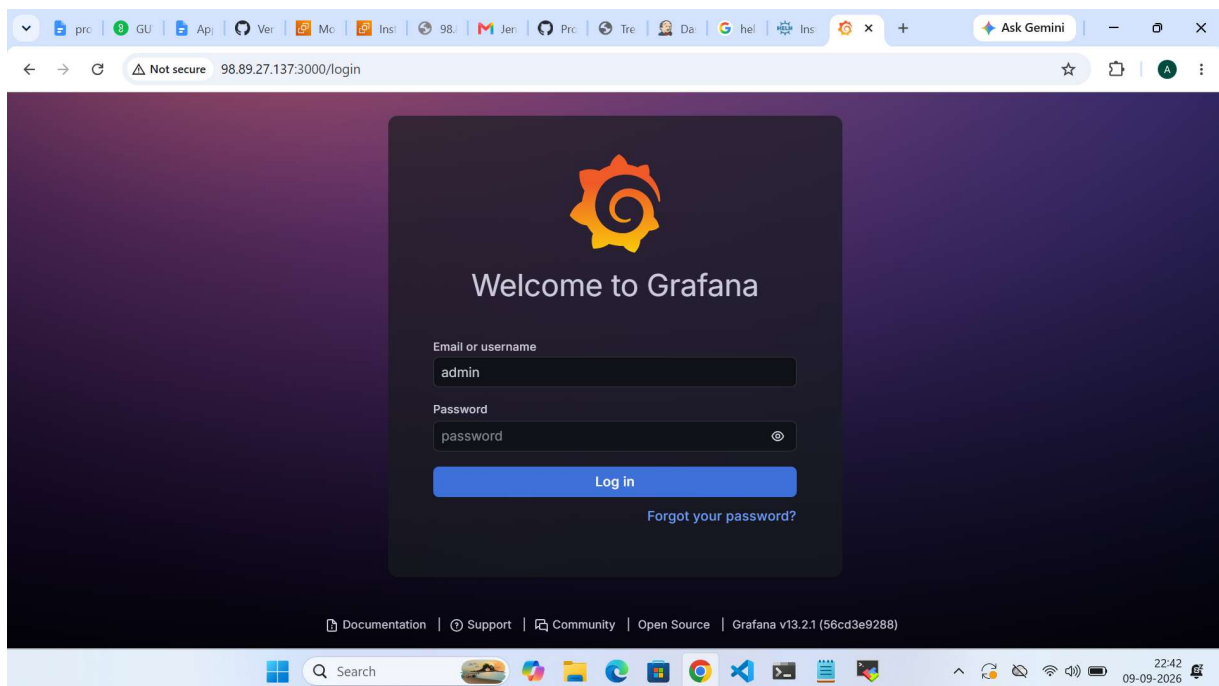
```
kubectl port-forward \
--address 0.0.0.0 \
-n monitoring \
svc/prometheus-stack-grafana \
3000:80
```

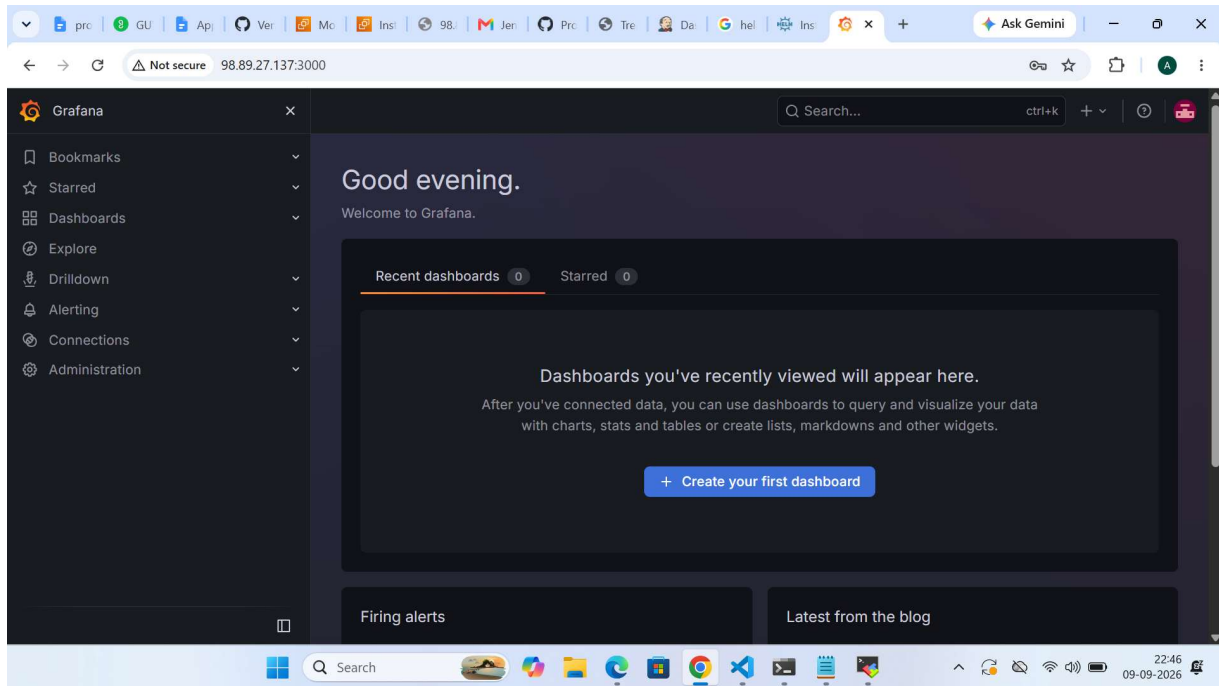
Login to grafna

Default User name: admin

Get password command

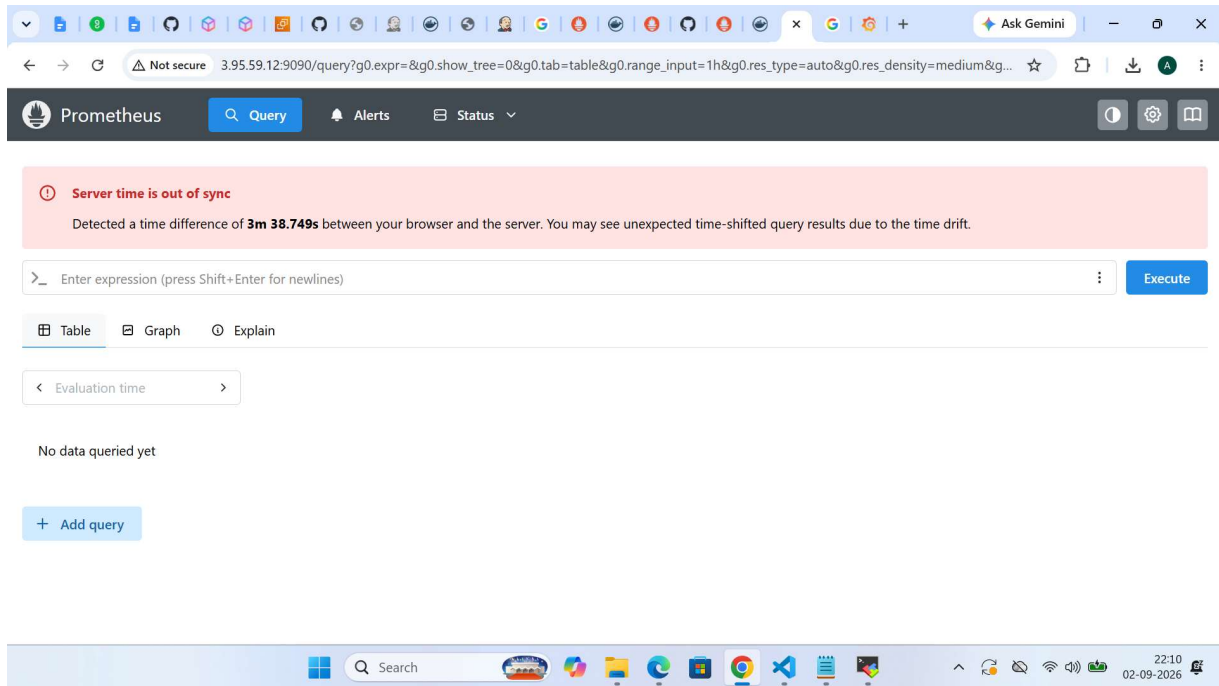
```
kubectl get secret \
-n monitoring \
prometheus-stack-grafana \
-o jsonpath="{.data.admin-password}" | base64 --decode
```





Verify prometheus

```
kubectl get svc -n monitoring | grep prometheus
kubectl port-forward \
  --address 0.0.0.0 \
  -n monitoring \
  svc/prometheus-stack-kube-prom-prometheus \
  9090:9090
```



Setting up alert manager

```
kubectl get svc -n monitoring | grep alertmanager
```

```
kubectl port-forward \  
--address 0.0.0.0 \  
-n monitoring \  
svc/prometheus-stack-kube-prom-alertmanager \  
9093:9093
```

Alertmanager Alerts Silences Status Settings Documentation [New Silence](#)

Filter Group Receiver: All ☐ Silenced ☐ Inhibited ☐ Muted

Custom matcher, e.g. `env="production"` [+](#) [Silence](#)

+ Expand all groups

+ null Not grouped 1 alert

+ null Not grouped 2 alerts

Monitoring Screenshots

Prometheus Query Alerts Status

`kube_deployment_status_replicas_available` [Execute](#)

Table Graph Explain

< Evaluation time > Load time: 370ms Result series: 6

<code>kube_deployment_status_replicas_available{container="kube-state-metrics", deployment="prometheus-stack-kube-prom-operator", endpoint="http", instance="192.168.33.202:8080", job="kube-state-metrics", namespace="monitoring", pod="prometheus-stack-kube-state-metrics-6ddf4f88fb-vltw7", service="prometheus-stack-kube-state-metrics"}</code>	1
<code>kube_deployment_status_replicas_available{container="kube-state-metrics", deployment="prometheus-stack-grafana", endpoint="http", instance="192.168.33.202:8080", job="kube-state-metrics", namespace="monitoring", pod="prometheus-stack-kube-state-metrics-6ddf4f88fb-vltw7", service="prometheus-stack-kube-state-metrics"}</code>	1
<code>kube_deployment_status_replicas_available{container="kube-state-metrics", deployment="prometheus-stack-kube-state-metrics", endpoint="http", instance="192.168.33.202:8080", job="kube-state-metrics", namespace="monitoring", pod="prometheus-stack-kube-state-metrics-6ddf4f88fb-vltw7", service="prometheus-stack-kube-state-metrics"}</code>	1
<code>kube_deployment_status_replicas_available{container="kube-state-metrics", deployment="trend-react-app", endpoint="http", instance="192.168.33.202:8080", job="kube-state-metrics", namespace="default", pod="prometheus-stack-kube-state-metrics-6ddf4f88fb-vltw7", service="prometheus-stack-kube-state-metrics"}</code>	2
<code>kube_deployment_status_replicas_available{container="kube-state-metrics", deployment="coredns", endpoint="http", instance="192.168.33.202:8080", job="kube-state-metrics", namespace="kube-system", pod="prometheus-stack-kube-state-metrics-6ddf4f88fb-vltw7", service="prometheus-stack-kube-state-metrics"}</code>	2
<code>kube_deployment_status_replicas_available{container="kube-state-metrics", deployment="metrics-server", endpoint="http", instance="192.168.33.202:8080", job="kube-state-metrics", namespace="kube-system", pod="prometheus-stack-kube-state-metrics-6ddf4f88fb-vltw7", service="prometheus-stack-kube-state-metrics"}</code>	2

Browser tabs: pri, GL, Ap, Ve, Se, In, 98, M, Je, Pr, Tr, D, G, he, In, D, x, +, Ask Gemini

Not secure 98.89.27.137:9090/query?g0.expr=kube_deployment_status_replicas_available%7B%0A++namespace%3D"default"%2C%0A++deployment...

Prometheus Query Alerts Status

Server time is out of sync
Detected a time difference of 3m 43.262s between your browser and the server. You may see unexpected time-shifted query results due to the time drift.

```
kube_deployment_status_replicas_available{
  namespace="default",
  deployment="trend-react-app"
}
```

Execute

Table Graph Explain

Evaluation time Load time: 338ms Result series: 1

```
kube_deployment_status_replicas_available{container="kube-state-metrics", deployment="trend-react-app", endpoint="http", instance="192.168.33.202:8080", job="kube-state-metrics", namespace="default", pod="prometheus-stack-kube-state-metrics-6dd4f88fb-vltw7", service="prometheus-stack-kube-state-metrics"}
```

+ Add query

Windows taskbar: Search, icons, 23:16, 09-09-2026

Browser tabs: pri, GL, Ap, Ve, Se, In, 98, M, Je, Pr, Tr, D, G, he, In, D, Gr, x, +, Ask Gemini

Not secure 98.89.27.137:9090/targets

Prometheus Query Alerts Status > Target health

serviceMonitor/monitoring/prometheus-stack-kube-prom-alertmanager/1 1 / 1 up

Endpoint	Labels	Last scrape	State
http://192.168.62.30:8080/metrics	container="config-reloader" endpoint="reloader-web" instance="192.168.62.30:8080" job="prometheus-stack-kube-prom-alertmanager" namespace="monitoring" pod="alertmanager-prometheus-stack-kube-prom-alertmanager-0" service="prometheus-stack-kube-prom-alertmanager"	4m 912ms ago 2ms	UP

serviceMonitor/monitoring/prometheus-stack-kube-prom-apiserver/0 2 / 2 up

Endpoint	Labels	Last scrape	State
https://192.168.116.170/metrics	endpoint="https" instance="192.168.116.170:443" job="apiserver" namespace="default" service="kubernetes"	3m 57.81s ago 137ms	UP
https://192.168.64.240/metrics	endpoint="https" instance="192.168.64.240:443" job="apiserver" namespace="default" service="kubernetes"	3m 57.292s ago 148ms	UP

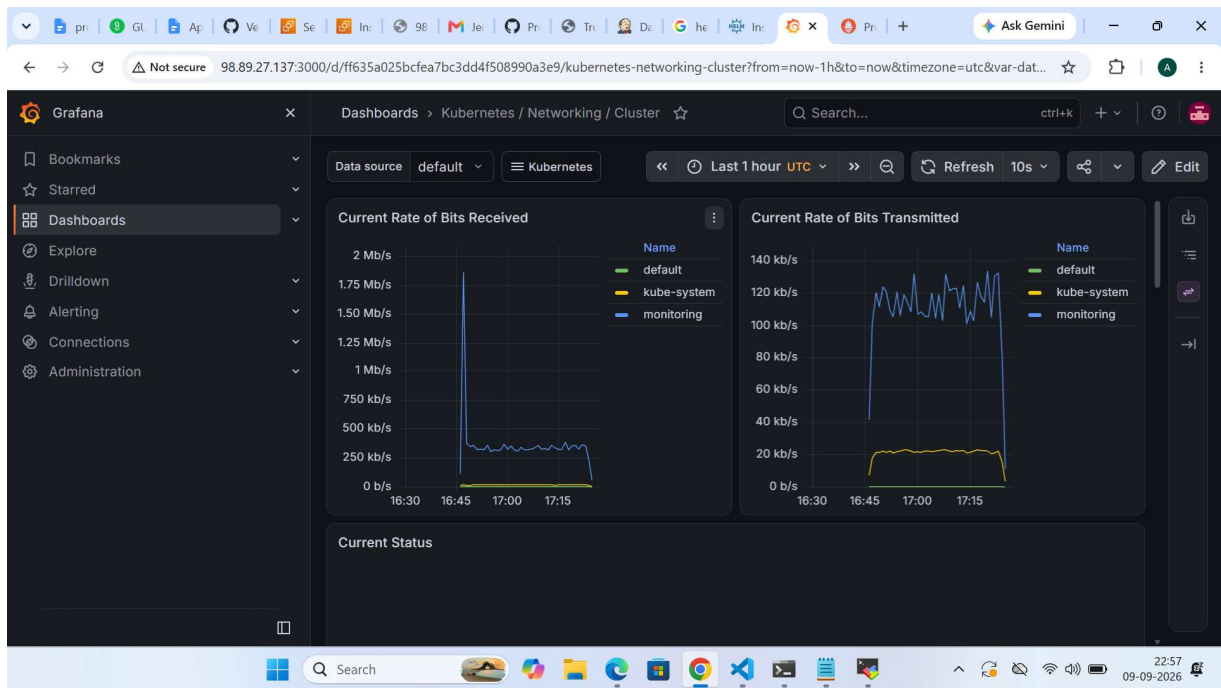
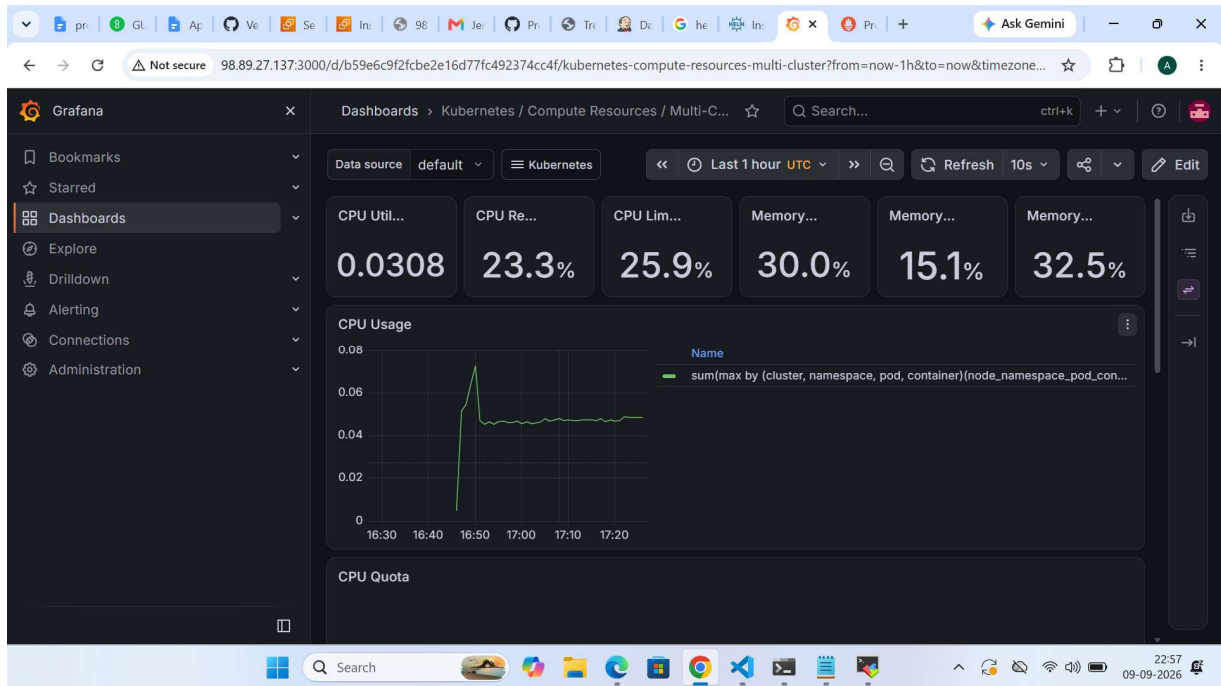
Windows taskbar: Search, icons, 22:54, 09-09-2026

Browser window showing Prometheus status page. The URL is `98.89.27.137:9090/targets`. The page displays two target health sections:

- serviceMonitor/monitoring/prometheus-stack-kube-prom-kube-proxy/0**: 2 / 2 up. Two endpoints are listed, both with a state of **UP**.
 - Endpoint: `http://192.168.42.45:10249/metrics`. Labels: `endpoint="http-metrics"`, `instance="192.168.42.45:10249"`, `job="kube-proxy"`, `namespace="kube-system"`, `pod="kube-proxy-6w12z"`, `service="prometheus-stack-kube-prom-kube-proxy"`. Last scrape: 3m 44.692s ago, 6ms.
 - Endpoint: `http://192.168.23.192:10249/metrics`. Labels: `endpoint="http-metrics"`, `instance="192.168.23.192:10249"`, `job="kube-proxy"`, `namespace="kube-system"`, `pod="kube-proxy-fmcdt"`, `service="prometheus-stack-kube-prom-kube-proxy"`. Last scrape: 3m 50.798s ago, 7ms.
- serviceMonitor/monitoring/prometheus-stack-kube-prom-kubelet/0**: 2 / 2 up. One endpoint is listed with a state of **UP**.
 - Endpoint: `https://192.168.23.192:10250/metrics`. Labels: `endpoint="https-metrics"`, `instance="192.168.23.192:10250"`, `job="kubelet"`, `metrics_path="/metrics"`, `namespace="kube-system"`, `node="ip-192-168-23-192.ec2.internal"`, `service="prometheus-stack-kube-prom-kubelet"`. Last scrape: 3m 48.706s ago, 29ms.

Browser window showing Grafana dashboard for Kubernetes / API server. The URL is `98.89.27.137:3000/d/09ec8aa1e996d6ffcd6817bbaff4db1b/kubernetes-api-server?from=now-1h&to=now&timezone=utc&var-datasource=...`. The dashboard displays the following metrics:

- Availability (30d) > 99.00...**: 100.000%
- Error Budget (30d) > 99.000%**: A line chart showing the error budget over time, with a green bar indicating the current value.
- Read Availabilit...**: 5 req/s, 200, 301
- Read SLI - Req...**: 10000%, 8000%
- Read SLI - Errors**: 10000%, Value
- Read SLI - Dura...**: 1 s, configmaps, customresour



98.89.27.137

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...

/home/ubuntu/

Name

- aws
- cache
- config
- gnupg
- kube
- ssh
- .bash_history
- .bash_logout
- .bashrc
- .profile
- .Xauthority

Remote monitoring

Follow terminal folder

```

ubuntu@ip-10-0-1-179:~$ helm list -n monitoring
NAME                NAMESPACE    REVISION    UPDATED                               STATUS    CHART
prometheus-stack    monitoring    1           2026-09-09 16:45:00.645513215 +0000 UTC deployed kube-prometheus-stack-90.0.0
ubuntu@ip-10-0-1-179:~$

```

ip-10-0-1-179 1% 1.26 GB / 3.74 GB 0.01 Mb/s 0.00 Mb/s 140 min /: 28% /tmp: 1% /boot: 11% /boot/efi: 7%

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23:01 09-09-2026

98.89.27.137

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...

/home/ubuntu/

Name

- aws
- cache
- config
- gnupg
- kube
- ssh
- .bash_history
- .bash_logout
- .bashrc
- .profile
- .Xauthority

Remote monitoring

Follow terminal folder

```

ubuntu@ip-10-0-1-179:~$ kubectl get pods -n monitoring
NAME                                                    READY   STATUS    RESTARTS   AGE
alertmanager-prometheus-stack-kube-prom-alertmanager-0 2/2     Running   0           41m
prometheus-prometheus-stack-kube-prom-prometheus-0     2/2     Running   0           41m
prometheus-stack-grafana-5ff9f59c99-d9kxp              3/3     Running   0           41m
prometheus-stack-kube-prom-operator-76d464f594-64tvc    1/1     Running   0           41m
prometheus-stack-kube-state-metrics-6ddf4f88fb-vltw7    1/1     Running   0           41m
prometheus-stack-prometheus-node-exporter-6jgwb        1/1     Running   0           41m
prometheus-stack-prometheus-node-exporter-pmv5p        1/1     Running   0           41m
ubuntu@ip-10-0-1-179:~$

```

ip-10-0-1-179 0% 1.25 GB / 3.74 GB 0.01 Mb/s 0.00 Mb/s 139 min /: 28% /tmp: 1% /boot: 11% /boot/efi: 7%

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23:00 09-09-2026

98.89.27.137

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...

/home/ubuntu/

Name

- ...
- .aws
- .cache
- .config
- .gnupg
- .kube
- .ssh
- .bash_history
- .bash_logout
- .bashrc
- .profile
- .Xauthority

Remote monitoring

Follow terminal folder

```
ubuntu@ip-10-0-1-179:~$ kubectl get svc -n monitoring
```

NAME	AGE	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)
alertmanager-operated	42m	ClusterIP	None	<none>	9093/TCP,
9094/TCP,9094/UDP					
prometheus-operated	42m	ClusterIP	None	<none>	9090/TCP
prometheus-stack-grafana	42m	ClusterIP	10.100.198.168	<none>	80/TCP
prometheus-stack-kube-prom-alertmanager	42m	ClusterIP	10.100.59.158	<none>	9093/TCP,
8080/TCP					
prometheus-stack-kube-prom-operator	42m	ClusterIP	10.100.180.227	<none>	443/TCP
prometheus-stack-kube-prom-prometheus	42m	ClusterIP	10.100.102.142	<none>	9090/TCP,
8080/TCP					
prometheus-stack-kube-state-metrics	42m	ClusterIP	10.100.220.111	<none>	8080/TCP
prometheus-stack-prometheus-node-exporter	42m	ClusterIP	10.100.91.69	<none>	9100/TCP

ubuntu@ip-10-0-1-179:~\$

ip-10-0-1-179 0% 1.25 GB / 3.74 GB 0.01 Mb/s 0.00 Mb/s 140 min /: 28% /tmp: 1% /boot: 11% /boot/efi: 7%

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23:01 09-09-2026